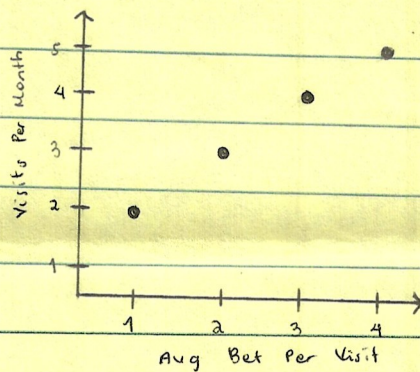


Casino Customers Dataset

Customer	Avg Bet per Visit	Visits per Month
A	1	2
B	2	3
C	3	4
D	4	5

① Scatter Plot



100
100

② Mean of each feature

$$\text{Mean of } X_1 = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

$$\text{Mean of } X_2 = \frac{2 + 3 + 4 + 5}{4} = 3.5$$

③ Center dataset using mean

$$A_{\text{center}} = \begin{bmatrix} 1 - 2.5 & 2 - 3.5 \\ 2 - 2.5 & 3 - 3.5 \\ 3 - 2.5 & 4 - 3.5 \\ 4 - 2.5 & 5 - 3.5 \end{bmatrix} = \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

④ Centered Data Matrix

⑤ Sample Covariance Matrix of the Centered Data

$$\Sigma = \frac{1}{n-1} A_{\text{center}}^T A_{\text{center}}$$

$$\Sigma = \frac{1}{4-1} \begin{bmatrix} -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \end{bmatrix} \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 2.25 + .25 + .25 + 2.25 & 2.25 + .25 \\ 2.25 + .25 + .25 + 2.25 & .25 + .25 \\ 2.25 & 2.25 \end{bmatrix}$$

$$A_{\text{center}} = \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

$$A_{\text{center}}^T = \begin{bmatrix} -1.5 & -0.5 & 0.5 & 0.5 \\ -1.5 & -0.5 & 1.5 & 1.5 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \Sigma = \begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix}$$

⑥ Eigenvalues of the covariance matrix.

$$\det(A - \lambda I) = 0$$

$$\det \left(\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$$

$$\det \begin{bmatrix} \frac{5}{3} - \lambda & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} - \lambda \end{bmatrix} = \lambda^2 + \frac{10}{3}\lambda = 0$$

$$= (\lambda + 0) \left(\lambda + \frac{10}{3} \right)$$

$$= \left(\frac{5}{3} - \lambda \right)^2 - \left(\frac{5}{3} \right)^2$$

$$= \left(\frac{5}{3} - \lambda \right) \left(\frac{5}{3} - \lambda \right) - \left(\frac{5}{3} \right)^2$$

$$\lambda_1 = \frac{10}{3} \quad \lambda_2 = 0$$

$$\det \begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0$$

$$= \left(\frac{5}{3} \right)^2 + 2 \left(\frac{5}{3} \lambda \right) + \lambda^2 - \left(\frac{5}{3} \right)^2$$

$$= \frac{25}{9} + \frac{10}{3} \lambda + \lambda^2 - \frac{25}{9} = 0$$

(7) Corresponding Eigenvectors

$$(A - \lambda I)$$

$$\lambda = \frac{10}{3}$$

$$A = \begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix}$$

$$\left(\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} - \begin{bmatrix} \frac{10}{3} & 0 \\ 0 & \frac{10}{3} \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{5}{3} - \frac{10}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} - \frac{10}{3} \end{bmatrix} = \begin{bmatrix} -\frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & -\frac{5}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\frac{5}{3}v_1 + \frac{5}{3}v_2 \\ \frac{5}{3}v_1 - \frac{5}{3}v_2 \end{bmatrix} = \begin{bmatrix} \frac{5}{3}v_2 = \frac{5}{3}v_1 \\ \frac{5}{3}v_1 = \frac{5}{3}v_2 \end{bmatrix}$$

$$\rightarrow v = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \checkmark$$

(8) Normalize the eigenvector

$$\lambda_1 = \frac{10}{3} \quad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \|v\| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

$$v_{\text{normal}} = \frac{v}{\|v\|} = \frac{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\sqrt{2}} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad u = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \checkmark$$

(9) Check if eigenvector and eigenvalues are correct

$$Av = \lambda v$$

$$\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{10}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{10}{3} \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{10}{3} \end{bmatrix} \checkmark$$

(10) $\frac{10}{3}$ is what I chose because of its higher variance from the start. $\checkmark$

(11) Project the centered data onto the first principal component

$$\text{New Projection} = A_{\text{centered}} * u$$

$$\begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$-\frac{1.5 - 1.5}{\sqrt{2}} = \frac{-3}{\sqrt{2}} \checkmark \quad \frac{0.5 + 0.5}{\sqrt{2}} = \frac{1}{\sqrt{2}} \checkmark$$

$$-\frac{0.5 - 0.5}{\sqrt{2}} = \frac{-1}{\sqrt{2}} \checkmark \quad \frac{1.5 + 1.5}{\sqrt{2}} = \frac{3}{\sqrt{2}} \checkmark$$

(12) Resulting 1-dimensional representation of each sample

$$= \begin{bmatrix} -\frac{3}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} \end{bmatrix} \checkmark$$